OCR Further Mathematics
Pure Core - Proof
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245

The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		$\mathbf{A}^2 = \begin{pmatrix} 4 & 12 \\ 0 & 4 \end{pmatrix}$, $\mathbf{A}^3 = \begin{pmatrix} 8 & 36 \\ 0 & 8 \end{pmatrix}$, $\mathbf{A}^4 = \begin{pmatrix} 16 & 96 \\ 0 & 16 \end{pmatrix}$ Conjecture: $\mathbf{A}^n = \begin{pmatrix} 2^n & 3n \times 2^{n-1} \\ 0 & 2^n \end{pmatrix}$	B1 (AO2.2a) B1 (AO2.2b) [2]	BC Allow this mark for any conjecture which works for $n = 1, 2, 3$ and 4.	
	b		Basis case: $n = 1$: $\mathbf{A}^1 = \begin{pmatrix} 2^1 & 3\times1\times2^0 \\ 0 & 2^1 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix} = \mathbf{A}$ so true for $n = 1$ Assume true for $n = k$ ie $\mathbf{A}^k = \begin{pmatrix} 2^k & 3k \times 2^{k-1} \\ 0 & 2^k \end{pmatrix}$ $\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A} = \begin{pmatrix} 2^k & 3k \times 2^{k-1} \\ 0 & 2^k \end{pmatrix}\begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix}$ $= \begin{pmatrix} 2^{k+1} & 3\times2^k + 3k\times2^k \\ 0 & 2^{k+1} \end{pmatrix}$ $= \begin{pmatrix} 2^{k+1} & 3(k+1)\times2^k \\ 0 & 2^{k+1} \end{pmatrix}$ So true for $n = k \Rightarrow$ true for $n = k + 1$. But true for $n = 1$. So true for all positive integer n	B1 (AO2.1) M1 (AO2.1) M1 (AO2.2a) A1 (AO2.4) [4]	Allow this mark even if the conjecture is wrong, provided that it works for $n = 1$ Must have statement in terms of some other variable than n. Conjecture need not be correct. Uses inductive hypothesis properly & expands AG.	A formal proof by induction is required for full marks.

Question			Answer/Indicative content	Marks	Part marks and guidance		
					Manipulating terms correctly and convincingly to obtain required form. Some intermediate working must be seen and a clear conclusion must be given for the induction process.		
			Total	6			

Question			Answer/Indicative content	Marks	Part marks and guidance	
2			Basis case: $1^3 + 2^3 + 3^3 = 36 = 4 \times 9$	B1 (AO2.1)	Basis case – either 4×9 or "36 is divisible by 9" given explicitly.	
			Consider the sum $f(r) = r^3 + (r + 1)^3 + (r + 2)^3$ Assume that $f(r) = 9k$ for some $k \in Z$ Then $f(r + 1) = f(r) + (r + 3)^3 - r^3$ $= f(r) + r^3 + 3r^2 \times 3 + 3r \times 9 + 27 - r^3$ $= f(r) + 9r^2 + 27r + 27$ $= 9k + 9(r^2 + 3r + 3) = 9k'$ for some $k' \in Z$ So if true for r then true also for $r + 1$ But it is true for $r = 1$ so is true for all integers r	M1 (AO2.1) M1 AO2.1) A1 (AO2.2a) A1 (AO2.5) [5]	For getting started (Sight of $f(r)$ is not necessary) For finding $f(r + 1)$ Each term as a multiple of 9 Conclusion dependant on all other marks being earned Examiner's Comments Many candidates completed this question successfully. However, a significant number of candidates dropped marks with the formal statements required for an inductive proof, either in the initial base case or in the final conclusion.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
3			Base case: For $n = 1$, $7^n + 3^{n-1} = 7+1 = 8$ which is a mulitple of 4. Assume that $f(k) = 7^k + 3^{k-1} = 4\lambda$ for some integer, λ $f(k+1) = 7^{k+1} + 3^k = 7.7^k + (7 - 4)3^{k-1}$ $= 7f(k) - 4.3^{k-1} = 7.4\lambda - 4.3^{k-1}$ $= 4(7\lambda - 3^{k-1}) = 4\lambda'$ Where λ' is an integer because λ is and so rhs is a multiple of 4. So if true for $n = k$, true also for $n = k + 1$. But it is true for $n = 1$ Therefore true for all positive integers	B1 (AO 2.1) M1 (AO 2.1) M1 (AO 2.1) A1 (AO 2.5) A1 (AO 2.2a) [5]	Alternatively: $f(k+1) = 7^{k+1} + 3^k = 7.7^k + 3.3^{k-1}$ $= 7(4k - 3^{k-1}) + 3.3^{k-1} = \ldots$	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4		a	$\mathbf{AB}=\begin{pmatrix}1&2\\a&-1\end{pmatrix}\begin{pmatrix}2&-1\\4&1\end{pmatrix}=\begin{pmatrix}10&1\\2a-4&-a-1\end{pmatrix}$ $(\mathbf{AB})\mathbf{C}=\begin{pmatrix}10&1\\2a-4&-a-1\end{pmatrix}\begin{pmatrix}5&0\\-2&2\end{pmatrix}$ $=\begin{pmatrix}48&2\\12a-18&-2a-2\end{pmatrix}$ $\mathbf{BC}=\begin{pmatrix}2&-1\\4&1\end{pmatrix}\begin{pmatrix}5&0\\-2&2\end{pmatrix}=\begin{pmatrix}12&-2\\18&2\end{pmatrix}$ $\mathbf{A}(\mathbf{BC})=\begin{pmatrix}1&2\\a&-1\end{pmatrix}\begin{pmatrix}12&-2\\18&2\end{pmatrix}$ $=\begin{pmatrix}48&2\\12a-18&-2a-2\end{pmatrix}=(\mathbf{AB})\mathbf{C}$ (which demonstrates associativity of matrix multiplication)	M1 (AO 3.1a) A1 (AO 2.1) M1 (AO 1.1) A1 (AO 2.1) [4]	Finding AB (or BC) Finding (AB)C (or A(BC)) Finding BC (or AB) Correct final matrix and statement of equality	
		b	$\mathbf{AC}=\begin{pmatrix}1&2\\a&-1\end{pmatrix}\begin{pmatrix}5&0\\-2&2\end{pmatrix}=\begin{pmatrix}1&4\\5a+2&-2\end{pmatrix}$ $\mathbf{CA}=\begin{pmatrix}5&0\\-2&2\end{pmatrix}\begin{pmatrix}1&2\\a&-1\end{pmatrix}=\begin{pmatrix}5&10\\2a-2&-6\end{pmatrix}\neq\mathbf{AC}$ (so matrix multiplication is not commutative)	M1 (AO 1.1) A1 (AO 2.1) [2]	Finding AC (or CA) Finding the other and statement of non-equality	
		c	$\begin{pmatrix}1&2\\a&-1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}x+2y\\ax-y\end{pmatrix}$ $x + 2y = 3x \Rightarrow y = x$ $ax - y = 3y$ and $y = x \Rightarrow a = 4$	M1 (AO 3.1a) A1 (AO 2.2a) A1 (AO 2.2a) [3]	Multiplying the vector into the matrix using the correct procedure	
			Total	9		

Question			Answer/Indicative content	Marks	Part marks and guidance
5			Base case, $n = 1$,	B1 (AO2.1)	
			$a_1 = \dfrac{11 \times 5^0 + 1}{2} = 6$ Assume true for $n = k$	M1 (AO2.1)	Statement of inductive hypothesis
			$\Rightarrow a_k = \dfrac{11 \times 5^{k-1} + 1}{2}$	M1 (AO2.2a)	Use of both recurrence relation for a_{k+1} in terms of a_k and also inductive hypothesis in attempt to derive required form for a_{k+1}
			Then $a_{k+1} = 5a_k - 2 = 5\left(\dfrac{11 \times 5^{k-1} + 1}{2}\right) - 2$ $= \left(\dfrac{11 \times 5^k + 5}{2}\right) - 2 = \left(\dfrac{11 \times 5^k + 1 + 4}{2}\right) - 2$ $= \left(\dfrac{11 \times 5^k + 1}{2}\right) + 2 - 2 = \dfrac{11 \times 5^{(k+1)-1} + 1}{2}$	A1 (AO2.5)	
			So if true for $n = k$ then true also for $n = k + 1$.	E1 (AO2.4)	
			But since it is true for $n = 1$ then it must be true for all integers $n \geq 1$.	[5]	
			Total	5	

Question			Answer/Indicative content	Marks	Part marks and guidance	
6			True for $k = 1$, since $1^5 - 1 = 0$ which is divisible by 5	B1 (AO2.1)		
			Assume true for $n = k$, then $k^5 - k$ is divisble by 5	M1 (AO2.1)		
			Then $(k + 1)^5 - (k + 1)$			
			$= k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1 - k - 1$			
			$= (5k^4 + 10k^3 + 10k^2 + 5k) + k^5 - k$	A1 (AO2.5)		
			$= 5(k^4 + 2k^3 + 2k^2 + k) + k^5 - k$			
			So if divisible by 5 for $n = k$ then also for $n = k + 1$	A1 (AO2.2a)		
			True for $n = 1$, so true generally	[4]		
			Total	4		

Exam Papers Practice

Question			Answer/Indicative content	Marks	Part marks and guidance		
7			Formula is true for $n = 1$ since $\frac{1}{5^1} = \frac{1}{5}$ and $\frac{5-4\times1}{5} = \frac{1}{5}$	B1(AO 2.5)	Must include an arithmetic justification		
			Assume that the formula is true for $n = k$	M1(AO 3.1a)			
			Then $\sum_{r=1}^{k+1} \frac{5-4r}{5} = \frac{k}{5^k} + \frac{5-4(k+1)}{5^{k+1}}$	M1(AO 2.1)	Add extra term		
			$= \frac{5k+5-4(k+1)}{5^{k+1}}$	A1(AO 2.2a)	Manipulate		
			$= \frac{k+1}{5^{k+1}}$				
			So if true for $n = k$ then true also for $n = k + 1$ So true generally	E1(AO 2.4)	Needs conclusion		
				[5]			
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance	
8		i	6 27	B1	Obtain correct values	
		i	129	B1	Obtain 3rd correct value Examiner's Comments Most candidates found the first 3 terms correctly and deduced a correct value for (ii) from their values.	
		ii	3	B1ft	State a correct value	
		iii	$5^{n+1}+2^n$	B1	Correct expression for u_{n+1} seen	Any letter, usually k or n
		iii		M1	Attempt to factorise $u_{n+1} + u_n$	Must deal with powers of 5 and 2
		iii		A1	Obtain correct simplified answer	e. g. $6 \times 5^n + 3 \times 2^{n-1}$
		iii		A1	Clear explanation why u_{n+1} is divisible by 3	Not $u_{n+1} + u_n$ divisible by 3
		iii		B1	Clear statement of induction process Examiner's Comments Some candidates tried to prove directly that u_{n+1} has a divisor of 3, with limited success. Those who showed that $u_{n+1} + u_n$ has a divisor of 3, then failed to explain clearly how this led to u_{n+1} having a divisor of 3.	Provided other 4 marks earned
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9		i	$\frac{2}{3}$, $\frac{2}{5}$, $\frac{2}{7}$	B1	B1 × 3, Obtain 3 correct values	
		i		B1		
		i		B1	Justify given answer Examiner's Comments Most found the first two answers correctly, but too frequently no working was shown to justify the value for u_4.	
		ii	$\frac{2}{2n-1}$	M1	Fraction, in terms of *n*, with correct numerator or denominator	
		ii		A1	Obtain correct answer a.e.f. Examiner's Comments A correct expression, in terms of *n*, was usually seen, but a minority gave an expression in terms of u_n.	
		iii	$\frac{2}{2(n+1)-1}$	B1ft	Verify result true when n = 1, for their (ii), or n = 2, 3 or 4	
		iii		M1	Expression for u_{n+1} using recurrence relation in terms of *n* using their (ii)	
		iii		A1	Correct unsimplified answer	
		iii		A1	Correct answer in correct form	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii		B1	Specific statement of induction conclusion, previous 4 marks must be earned, $n = 1$ must be verified Examiner's Comments Sufficient working to demonstrate that the result was true for $n = 1$ was generally seen, as was the working to derive the correct expression for u_{k+1} from the recurrence relation and their part (ii). However, candidates should be made aware that the work required in an induction question is mainly of the "given answer" type, so sufficient justification is essential. Many candidates did not give a satisfactory conclusion to the induction process.	
			Total	10		

Exam Papers Practice

Question			Answer/Indicative content	Marks	Part marks and guidance	
10				B1	Show clearly result true for $n = 1$, accept $(u_1) = 2 \times 3 - 1 = 5$	
			$3(2 \times 3^n - 1) + 2$	M1	Substitute for u_n in recurrence relation	
				A1	Establish correct result for u_{n+1} convincingly	
				B1	Clear statement of induction conclusion, provided 1st 3 marks earned Examiner's Comments Some candidates failed to show clearly that the result is true for $n = 1$, while others started the process at $n = 2$. Most were able to show that using the given form in the recurrence relation set up the induction process correctly. A significant number of candidates did not give a clear explanation of the induction process, so the last mark was often lost.	
			Total	4		

Exam Papers Practice

Question			Answer/Indicative content	Marks	Part marks and guidance	
11				B1	Show sufficient working to verify result true when $n = 1$	
			$k(k+1)^2 + (k+1)(3k+4)$	M1*	Add next term in series	
				DM1	Attempt to factorise their expression	
			$(k+1)(k+2)^2$	A1	Sufficient working to obtain this correct answer	
				B1	Clear statement of induction process, provided previous 4 marks earned Examiner's Comments This was not done particularly well. Most could establish the truth of the result for $n = 1$, but then did not add on correct $(k+1)$th term, or omitted brackets and so obtained an expression that could not be simplified. Those who had a correct expression often showed insufficient working to justify the truth for $k + 1$. Many lost the final mark by not giving a clear statement of how induction works.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12				B1	Establish result true for n = 1 or n = 2	
				M1	Multiply M and M^k, either order	
			$2\left(2^{k+1}-2\right)+2$ or $2^{k+1}+2^{k+1}-2$	A1	Obtain correct element	
				A1	Obtain other 3 correct elements	
				A1	Obtain $2^{k+2}-2$ convincingly	
				B1	Specific statement of induction conclusion, provided 5/5 earned so far and verified for n = 1 Examiner's Comments This was one of the less well-attempted questions. A significant minority thought that the induction step required the addition of M^k and M. Many failed to show sufficient working either in establishing the validity when n = 1, or in obtaining the elements in M^{k+1}. Centres should remind candidates of this, as well as the need for a clear statement of the induction conclusion.	
			Total	6		